

CASE STUDY

Smart Factory Transformation — Automotive OEM

Case Study | Pune, Maharashtra | 2023

Executive Summary

A leading automotive OEM in Pune engaged us to transform three of its high-volume assembly plants into fully connected smart factories. The 18-month programme—spanning IoT sensor deployment, MES integration, SCADA modernisation, and real-time Power BI dashboards—delivered measurable operational improvements and a clear path to further Industry 4.0 maturity.

22%	15%	■ 12 Cr	18 mo
OEE Improvement	Downtime Reduction	Annual Savings	Programme Duration

Client Context

The client—a Tier-0.5 supplier producing 4-wheelers and light commercial vehicles—faced escalating quality escapes, unpredictable line stoppages, and limited production visibility. Legacy PLCs ran on isolated networks; shop-floor data existed only in paper shift logs. The leadership team set an ambitious OEE target of 85% across all lines within two years.

Scope of Work

- Deployment of 1,200+ IoT edge sensors (vibration, temperature, cycle-count) across all three plants
- MES rollout (SAP ME) integrated with existing ERP (SAP S/4HANA) and quality management systems
- SCADA layer upgrade to enable real-time OT data streaming to cloud data platform
- Power BI dashboards for plant managers, shift supervisors, and C-suite (12 distinct report views)
- Digital twin of Assembly Line 3 for process simulation and capacity planning
- Change management & operator training programme (400+ shop-floor staff upskilled)

Technology Stack

Layer	Technology	Vendor/Product
IoT Edge	Edge gateways + MQTT brokers	Azure IoT Edge
MES	Manufacturing Execution System	SAP ME 15.4
SCADA	Supervisory control layer	Wonderware InTouch
Analytics	Real-time dashboards	Microsoft Power BI

Cloud	Data lake & processing	Azure (ADX + ADF)
Integration	ERP-MES connector	Custom middleware (Python)

Key Outcomes

- Overall Equipment Effectiveness rose from 61% to 83%, exceeding the 85% target trajectory
- Unplanned downtime reduced by 15%; mean-time-to-repair improved by 28%
- Annual cost savings of ■12 Cr through reduced scrap, rework, and energy optimisation
- First-pass yield on critical sub-assemblies improved from 94.1% to 98.6%
- Real-time production visibility enabled proactive scheduling adjustments, reducing overtime by 22%

Our Differentiators

Our pre-built Industry 4.0 accelerator—comprising 40+ reusable integration adapters and a validated OT/IT convergence architecture—reduced the integration effort by an estimated 35% compared to a greenfield build. The automotive domain team (average 12 years shop-floor experience) earned trust with plant engineers rapidly, accelerating the change management phase.

Team & Timeline

- Programme team: 25 consultants (8 OT/IIoT engineers, 6 SAP specialists, 4 data engineers, 7 project leads)
- Phase 1 (Pilot — Line 1): 4 months
- Phase 2 (Scale — Lines 2 & 3): 10 months
- Phase 3 (Optimise & Handover): 4 months